

UG Project (EC22711) : 2026-2027			
Design of a Fault-Tolerant ARINC 429 Communication Controller ASIC			
Internal Project			
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ABSTRACT			
Aircraft communication systems require high reliability to ensure safe and continuous data exchange between avionics subsystems such as the Flight Management System (FMS), Air Data Computer (ADC), GPS, and cockpit displays. The ARINC 429 communication protocol is one of the most widely used standards in commercial aircraft for this purpose. Conventional ARINC 429 controllers mainly perform data transmission, reception, and parity error detection, but they cannot determine whether an error is temporary or persistent, which may affect the overall reliability of the communication system. This project proposes the Design and ASIC Implementation of a Health-Aware ARINC 429 Communication Controller. The proposed controller includes a hardware-based health monitoring unit that continuously monitors communication errors, timeout events, and fault history. Based on these observations, the controller classifies the communication health and performs adaptive fault recovery by selecting suitable actions such as normal operation, retransmission, controller reset, or fault reporting. The complete architecture is designed using Verilog HDL, verified through functional simulation and fault injection, and implemented on an FPGA. The design can also be extended to a complete RTL-to-GDSII ASIC flow using industry-standard or open-source VLSI tools. The project is carried out in two phases to provide a systematic development and evaluation process. Phase 1 focuses on the design, implementation, and verification of a conventional ARINC 429 communication controller, establishing a baseline for comparison. Phase 2 extends the controller by integrating a hardware-based health monitoring unit, fault-history tracking, fault classification, and adaptive recovery mechanisms to improve communication reliability and fault tolerance. The proposed design aims to enhance the robustness of ARINC 429 communication while maintaining low hardware resource utilization, low power consumption, and minimal timing overhead. Its performance will be evaluated against the conventional controller based on fault detection capability, recovery latency, FPGA resource utilization, timing performance, and power consumption. This phased approach demonstrates how advanced VLSI design techniques can improve aircraft communication systems through intelligent hardware-based health monitoring and adaptive fault recovery while quantifying the hardware overhead introduced by the additional functionality.			
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